

Methods and Computing

DoSS Summer Prep Bootcamp 2026

1 Time and Place

Exact times TBA

2 Instructor

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3 Course outline

This course will revisit some basic programming principles in R and review essential concepts in likelihood inference.

4 Textbooks

4.1 Primary Textbooks

- *All of Statistics* by L. Wasserman (AoS)
- *Statistical Inference, Second Edition* by George Casella and Robert L. Berger (C&B)

4.2 Optional Texts

- *Statistical Models* by A.C. Davison
- *Mathematical Statistics* by K. Knight
- *Theory of Point Estimation* by E.L. Lehmann

5 Tentative Lecture Schedule

The Lecture Topics and corresponding texts are outlined below. This schedule is tentative and may be changed as the course progresses.

Module	Topics	References
1	Tidyverse basics; data wrangling & graphing; Using LLMs to generate R code	-
2	Statistical inference (I)	AoS Chp 1-5; AoS Chp 6; C&B Chp 6.3, 7
3	Statistical inference (II)	AoS Chp 8; C&B Chp 8
4	Linear Regression & Generalized Linear Models	AoS Chp 13; C&B Chp 11-12
5	Simulation and parallel computing	C&B Chap 10; AoS Chp 5, 24;